MONITOR DEVICES Through Web Browser

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The Internet of Things (IoT) is being widely used in industrial and commercial applications. This article describes how to monitor things or devices connected to WeMos D1 board, which acts as a Web server. Pin description of WeMos D1 board is shown in Fig. 1.

Hypertext Transfer Protocol (HTTP) in the IoT works as a request and response protocol between a client and server. A Web browser functions as a client on a computer that requests information from the website (server). When you type the Web address in the Web browser, it acts as a client and requests contents of the Web page in HTML format from the website. The website, acting as a server, is always listening for such requests from the client (Web browser) and responds with contents of the page.

Circuit and working

The circuit diagram for monitoring things through a Web browser is shown in Fig. 2. It is built around WeMos D1 (Board1) and an analogue sensor (LM35). To monitor the sensor, it is necessary to connect it to analogue input pin of Board1.

However, we will first test the project without using the sensor. We should be able to see the float voltage value on the browser, that is, reading a random value on the browser without connecting the sensor to input pin A0 of WeMos D1. All this is monitored by the program (wemos_sensordata.ino) inside WeMos D1 board. Data read from pin A0 is sent as HTTP response to the client (Web browser).

Construction and testing

WeMos D1 board can be programmed using Arduino IDE. To use WeMos D1 board in Arduino IDE, install the board library in Arduino IDE, based on the steps given below.

1. Open Arduino IDE and paste the following address from File→Preferences→Additional Board Manager URLs:

http://arduino.esp8266.com/stable/package_esp8266com_index.json

2. Select ESP8266 by ESP8266 Community, and click on Install from Tools→Boards→Board Manager.

3. Close Board Manager. Go to Tools→Board and select WeMos D1 R2 & mini option.

4. Select 115200 from Tools→Upload Speed.

To connect WeMos D1 board to the computer, you may need a USB-to-serial chip driver. Install the driver from the following URL:

http://github.com/nodemcu/nodemcu-devkit/tree/master/Drivers

After connecting WeMos D1 board to the computer/laptop using a USB cable, open the source code (wemos_sensordata.ino) from Arduino IDE.

Before uploading the code to WeMos D1 board, make sure to replace the SSID and password in the program/sketch with your Wi-Fi username and password. Compile and upload the code to WeMos D1 board.

Open Serial Monitor to view the IP address of WeMos D1 board assigned

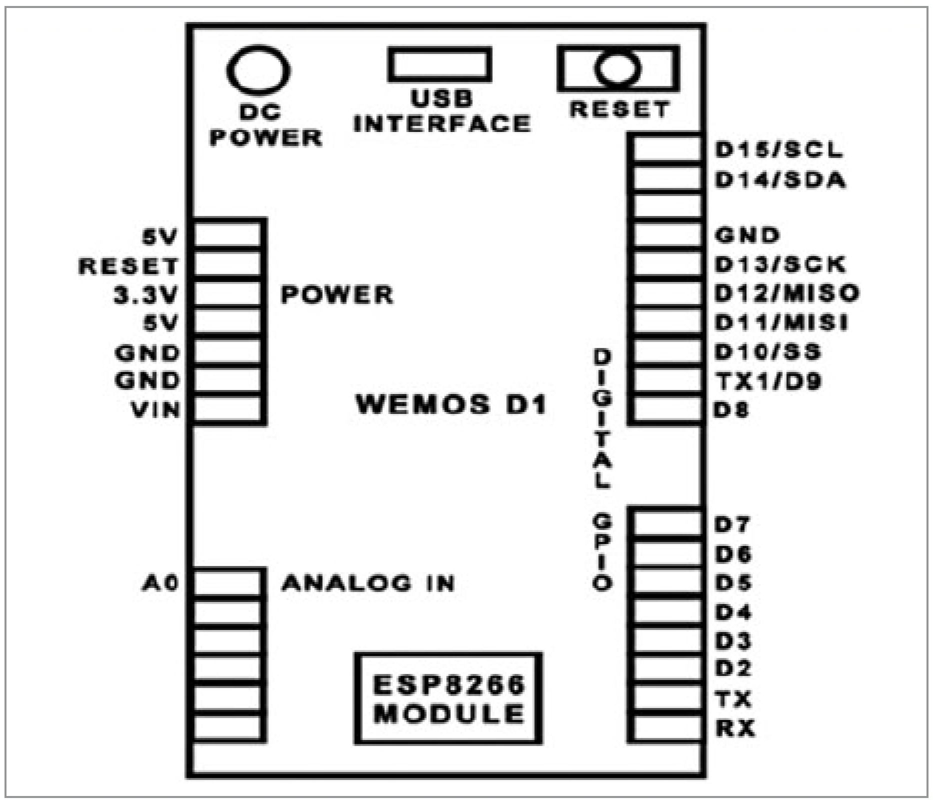

Fig. 1: WeMos D1 board pin description

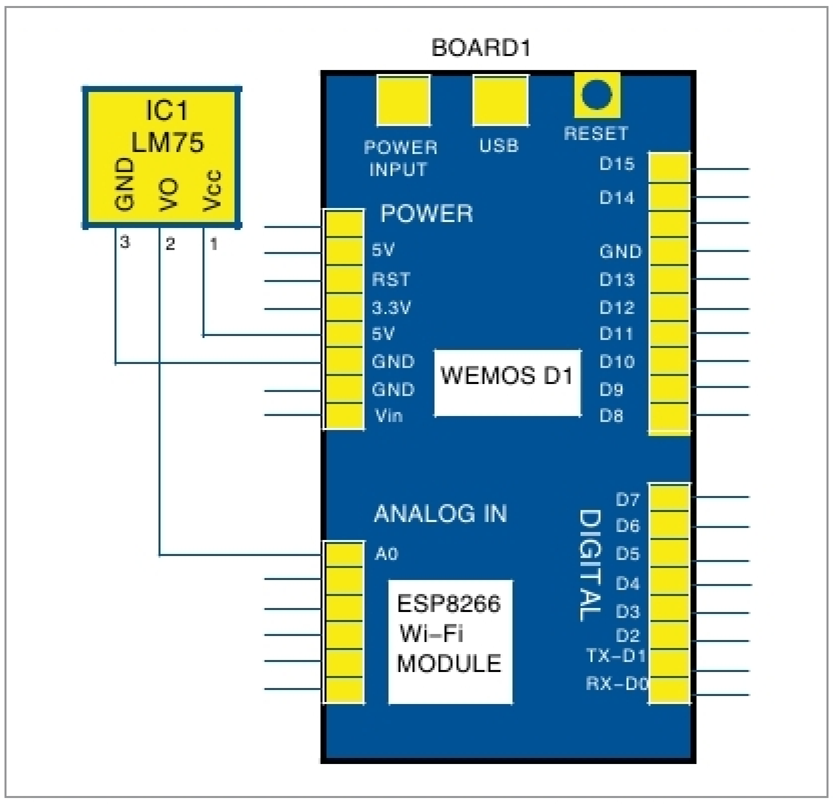

Fig. 2: Circuit diagram for monitoring devices through Web browser